Sample Papers

CS402

Final Term Examination – Spring 2006

Time Allowed: 150 Minutes

Question No. 1 **Marks : 1**

A production of the form non-terminal → non-terminal is called a dead Production.

1. True
2. False

Question No. 2 **Marks : 3**

Semi-word is a string having some terminals and one non-terminal at the right of string.

1. True
2. False

www.ilmkidunya.com

Question No. 3 **Marks : 3**

Two FAs are equivalent if they have same no. of states.

1. True
2. False

Question No. 4 **Marks : 1**

There exist exactly two different derivations in an ambiguous CFG for a word.

1. True
2. False

Question No. 5 **Marks : 1**

Construct RE's for following languages over Σ= {a, b}

i. All words which contain substring "baba". [5]

ii. All words which never end in substrings "aa" and "ab" [5]

Question No. 6 **Marks : 1**

Regular languages are closed under Union, Concatenation and Kleene star.

1. True
2. False

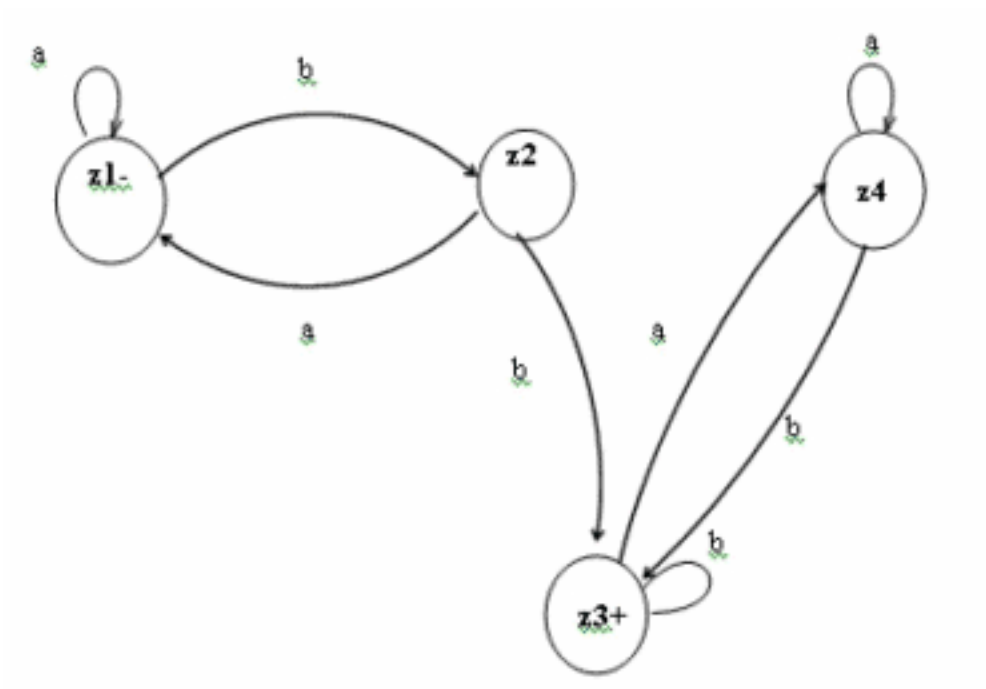
Question No. 7 **Marks : 1**

CFG may also represent a regular language.

1. True
2. False

Question No. 8 **Marks : 1**

Find the complement of the FA below,



Question No. 9 **Marks : 1**

PDA is stronger than FA.

3. True
4. False

Question No. 10 **Marks : 3**

Union of two FAs is an FA that accepts all the string of FA1 and FA2.

True
False

Question No. 11 **Marks : 3**

Given a CFG below,

a) Remove Unit productions from the CFG. [5]

$S \rightarrow S + T \mid T$
$T \rightarrow T * F \mid F$
$F \rightarrow (S) \mid a$

b) Derive CNF corresponding to the CFG given below. [10]

$S \rightarrow ABa$
$A \rightarrow aab$

Question No. 12 **Marks : 10**

In non-deterministic PDA a string may be traced by more than one paths.

1. True
2. False

Question No. 13 **Marks : 1**

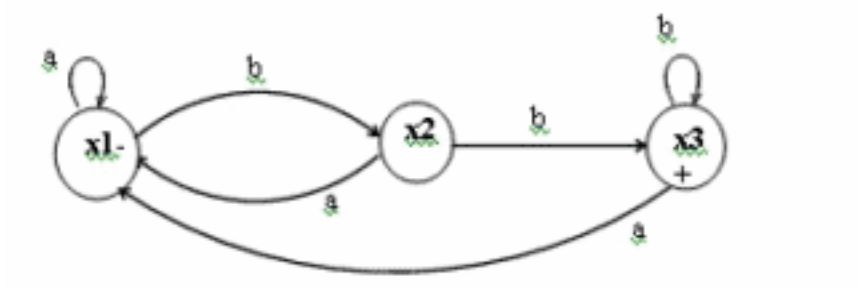
There always exist an FA for each PDA.

True
False

www.ilmkidunya.com

Question No. 14 **Marks : 1**

a) Given an FA below, [5]



What is language accepted by above FA?

b) Draw a non-deterministic PDA for the above FA. [10]